3 (a)(i)	As the space shuttle's **direction of velocity / motion is constantly changing**, the shuttle is accelerating. For the shuttle to **accelerate**, a resultant force must be acting on it according to **Newton's second law of motion**.	**B1** **B1**
(ii)	Gravitational force F_G provides centripetal force F_C. $F_G = F_C$ $GMm / (5R)^2 = mv^2 / 5R$ $v^2 = \underline{\frac{GM}{5R}}$	 **M1** **A1**
(iii)	$E_K = ½ mv^2$ $= ½ m (\frac{GM}{5R})$ $= \frac{GMm}{10R}$	 **M1**
(b)	Total Energy = KE + GPE $= \frac{GMm}{10R} + (-\frac{GMm}{5R})$ $= -\frac{GMm}{10R}$ $= – G(7.8 \times 10^{24})(9.0 \times 10^4) /(10 \times 6.8 \times 10^6)$ $= – 6.8858 \times 10^{11}$ J $= \underline{– 6.9 \times 10^{11} \text{ J}}$ (2 sf)	 **C1** **A1**
(c)(i)	At Z, $g_X = g_Y$ $GM/x^2 = G(2M) / (6.0 \times 10^8 – x)^2$ $(6.0 \times 10^8 – x)^2 / x^2 = 2$ $x = 2.4853 \times 10^8$ m $= \underline{2.5 \times 10^8 \text{ m}}$ (2 sf)	 **C1** **A1**
(ii)	At Z, GPE $= U_X + U_Y$ $= -\frac{GMm}{2.5\times10^8} + (-\frac{G(2M)m}{(6.0\text{-}2.5)\times10^8})$ $= –4.5486 \times 10^{11}$ J $= \underline{–4.5 \times 10^{11} \text{ J}}$ (2 sf)	 **C1** **A1**
(iii)	The space shuttle needs just enough energy to reach point Z, and the resultant gravitational force acting on it will be in the direction of planet Y. extra KE needed = work done in moving from X to Z $= [-4.5486 – (-6.8858)] \times 10^{11}$ J $= 2.3372 \times 10^{11}$ J $= \underline{2.3 \times 10^{11}}$ J (2 sf)	 **C1** **A1**